第十八章 磁力

- 核心概念
  - 洛仑兹力
  - 安培力
- 核心问题
  - 如何理解磁场和电场的相对性和统一性？
  - 什么是霍尔效应？
  - 安培力的本质是什么？
  - 磁矩在磁场中的力矩和势能？

第十八章 磁力

18.1 带电粒子在磁场中运动

18.2 电场和磁场的相对性和统一性

18.3 霍尔效应

18.4 安培力

18.5 载流线圈在磁场

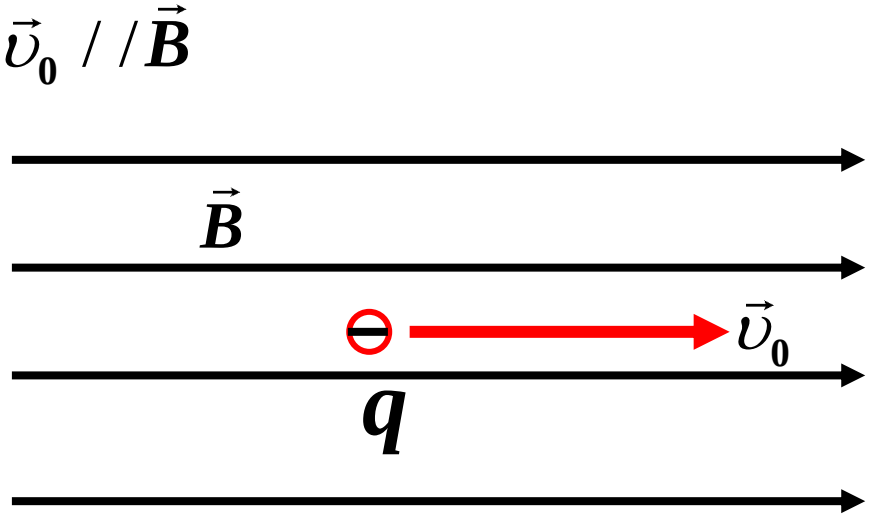
18.1 带电粒子在磁场中运动

洛仑兹力 $$\boxed{\vec{f}_m = q\vec{v} \times \vec{B}}$$

- 速度是相对于观察者所在参考系；
- 洛伦兹力不做功；
- 相对论条件下，牛顿第二定律形式不变

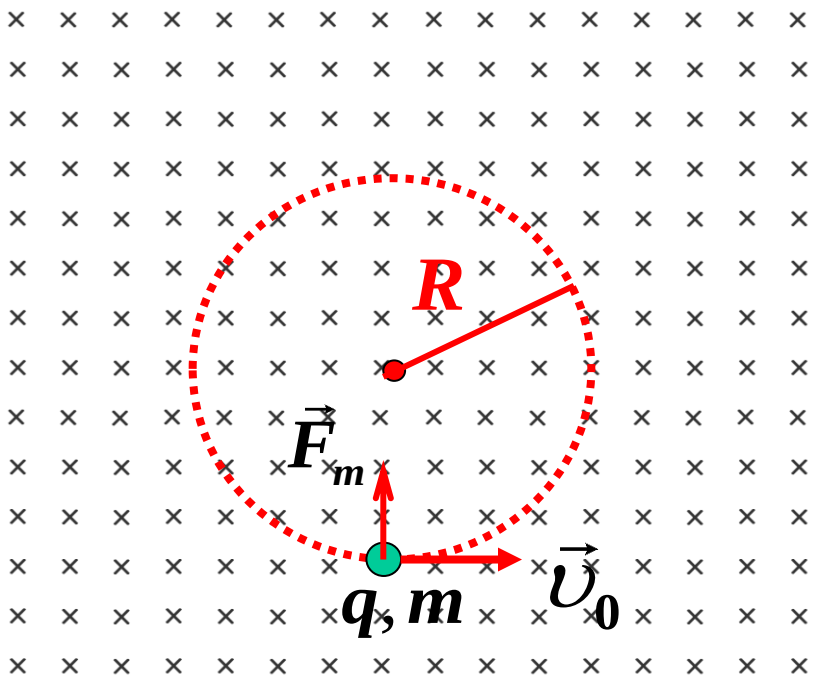
$$\vec{a} = \frac{\vec{F}}{m} - \frac{\vec{\upsilon}(\vec{\upsilon} \cdot \vec{F})}{mc^2} \qquad m = \frac{m_0}{(1 - \frac{\upsilon^2}{c^2})^{1/2}}$$

1、均匀磁场



沿磁场方向做匀速直线运动

$\vec{\upsilon}_0 \perp \vec{B}$



对称性原理　　在垂直于磁场的平面内运动

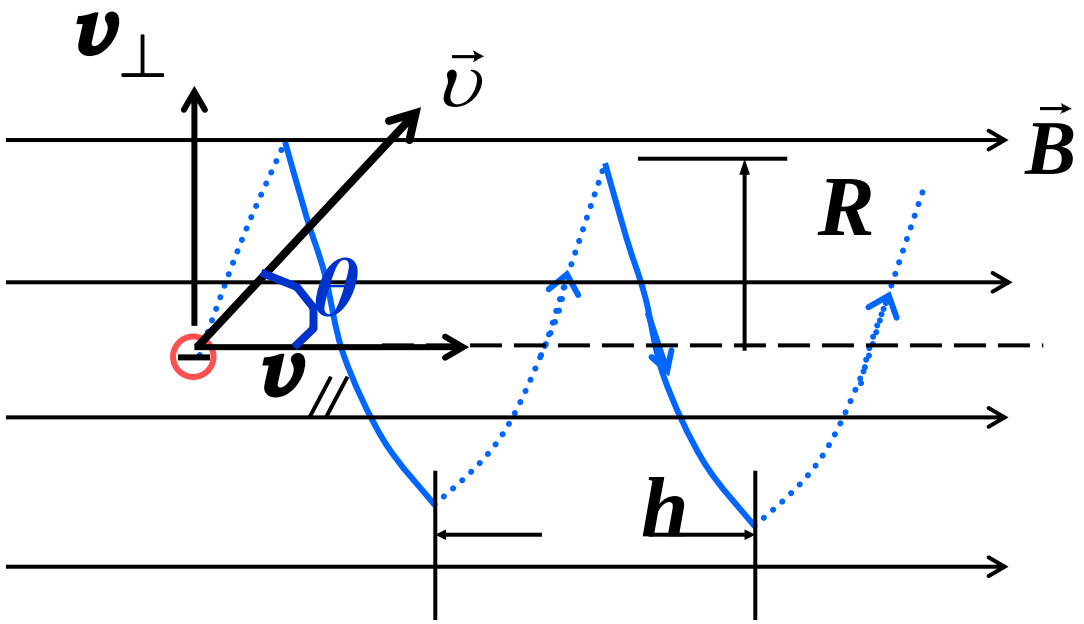
$\vec{\upsilon}_0 \perp \vec{F}$　　匀速率的圆周运动

$$q\upsilon B = m\upsilon^2 / R$$

半径　$R = m\upsilon / qB$

周期　$T = 2\pi R / \upsilon = 2\pi m / qB$　与速率无关

$\vec{\upsilon}_0$ 和 $\vec{B}$ 既不平行也不垂直



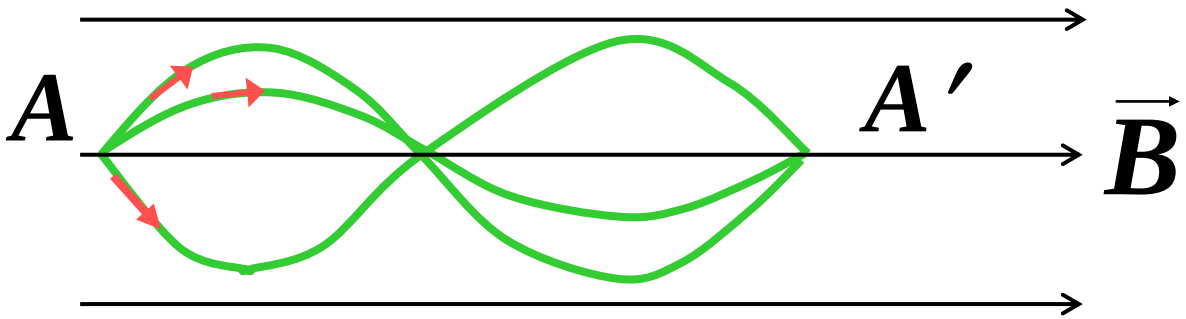
$$\upsilon_{//} = \upsilon\cos\theta \qquad \upsilon_\perp = \upsilon\sin\theta$$

轴线沿磁场方向的螺旋运动

$$R = m\upsilon_\perp / qB \qquad T = 2\pi m / qB$$

螺距 $h = \upsilon_{//}T = \dfrac{2\pi m}{qB}\upsilon_{//} = \dfrac{2\pi m}{qB}\upsilon\cos\theta$

磁聚焦



在A点 发散角不大，和速率几乎相同的条件下

$$\upsilon_{//} = \upsilon\cos\theta \qquad \upsilon_{\perp} = \upsilon\sin\theta$$

$$h = \upsilon_{//}T = \frac{2\pi m}{qB}\upsilon_{//} = \frac{2\pi m}{qB}\upsilon\cos\theta$$

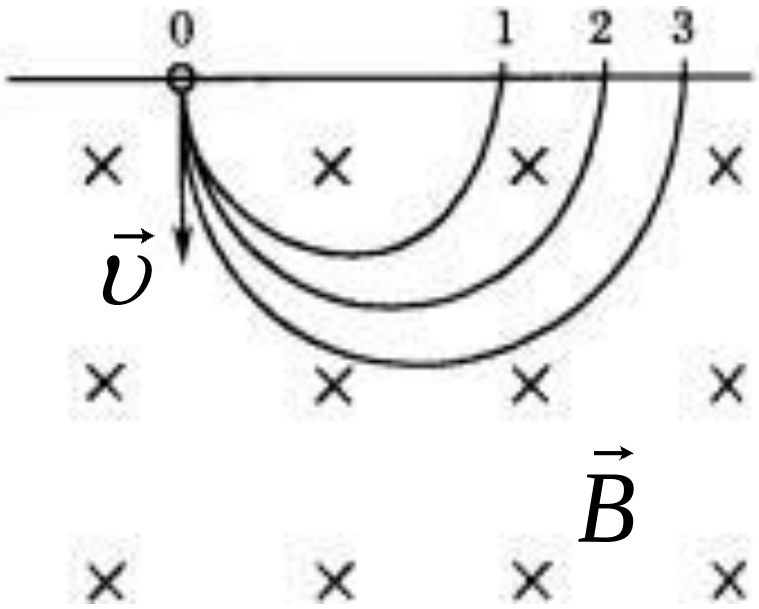
经过螺距的整数倍后相遇

质谱仪和动量谱仪

$$R = m\upsilon / qB$$

$$q / m = \upsilon / RB$$

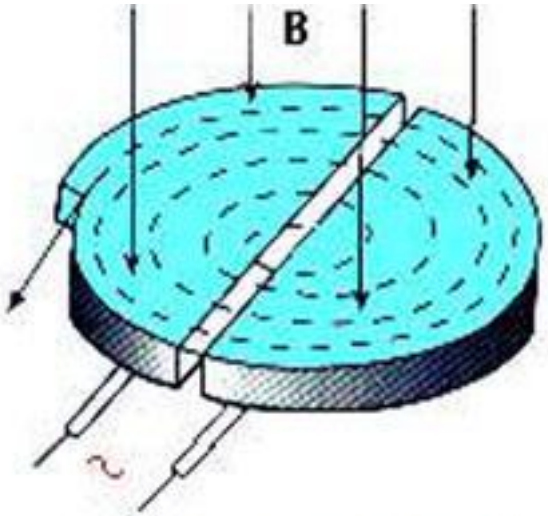
$$R = p / qB$$



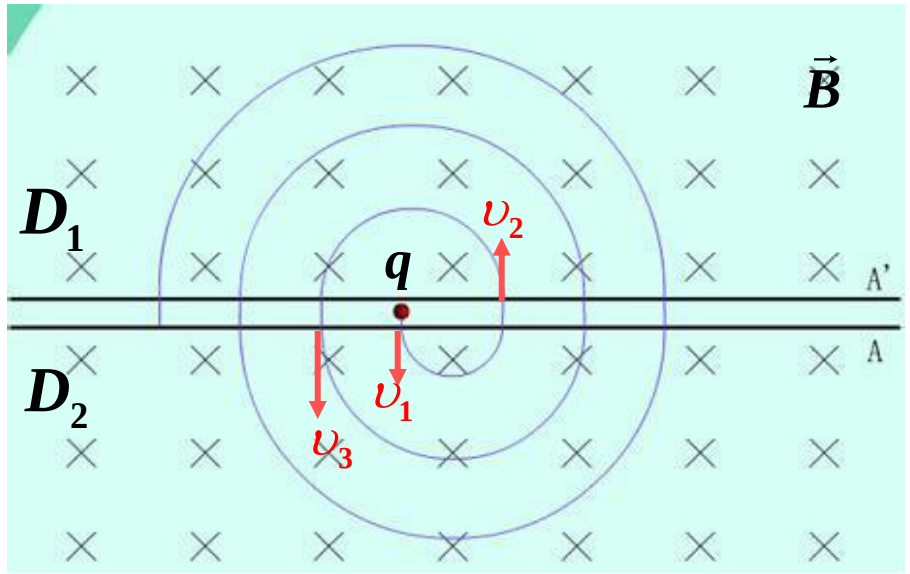
回旋加速器

$$T = 2\pi m / qB$$

周期与速率无关
（非相对论条件下）



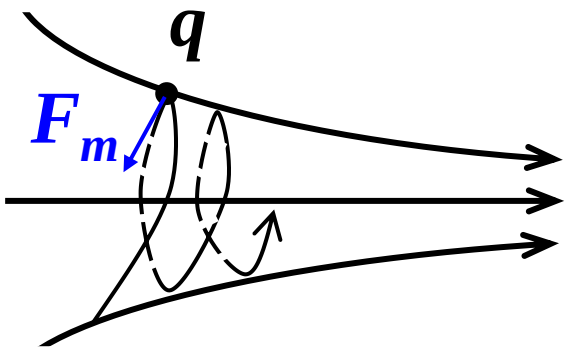
回旋加速器的D型盒



2、不均匀磁场

➢磁镜

强度逐渐增加的磁场能使粒子反射



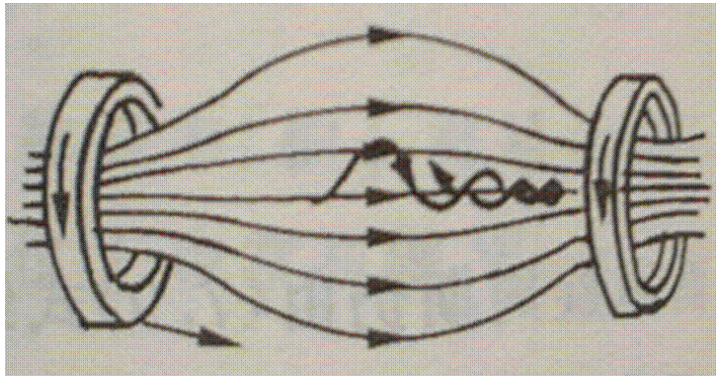
定性分析

洛伦兹力有与前进方向相反的分量

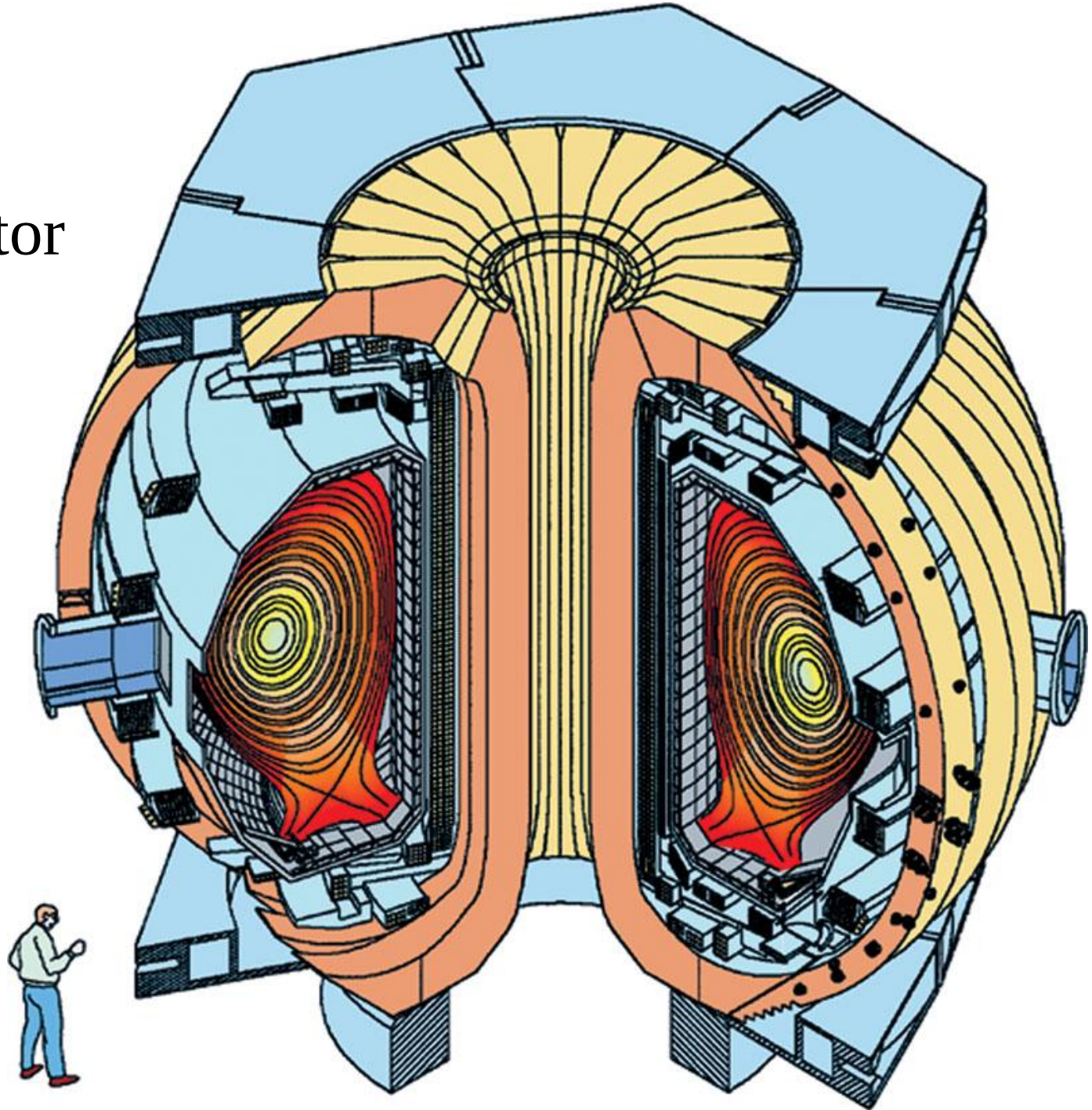
速率不变

▲磁约束

磁瓶



Tokamak
Fusion reactor



*绚丽多彩的极光

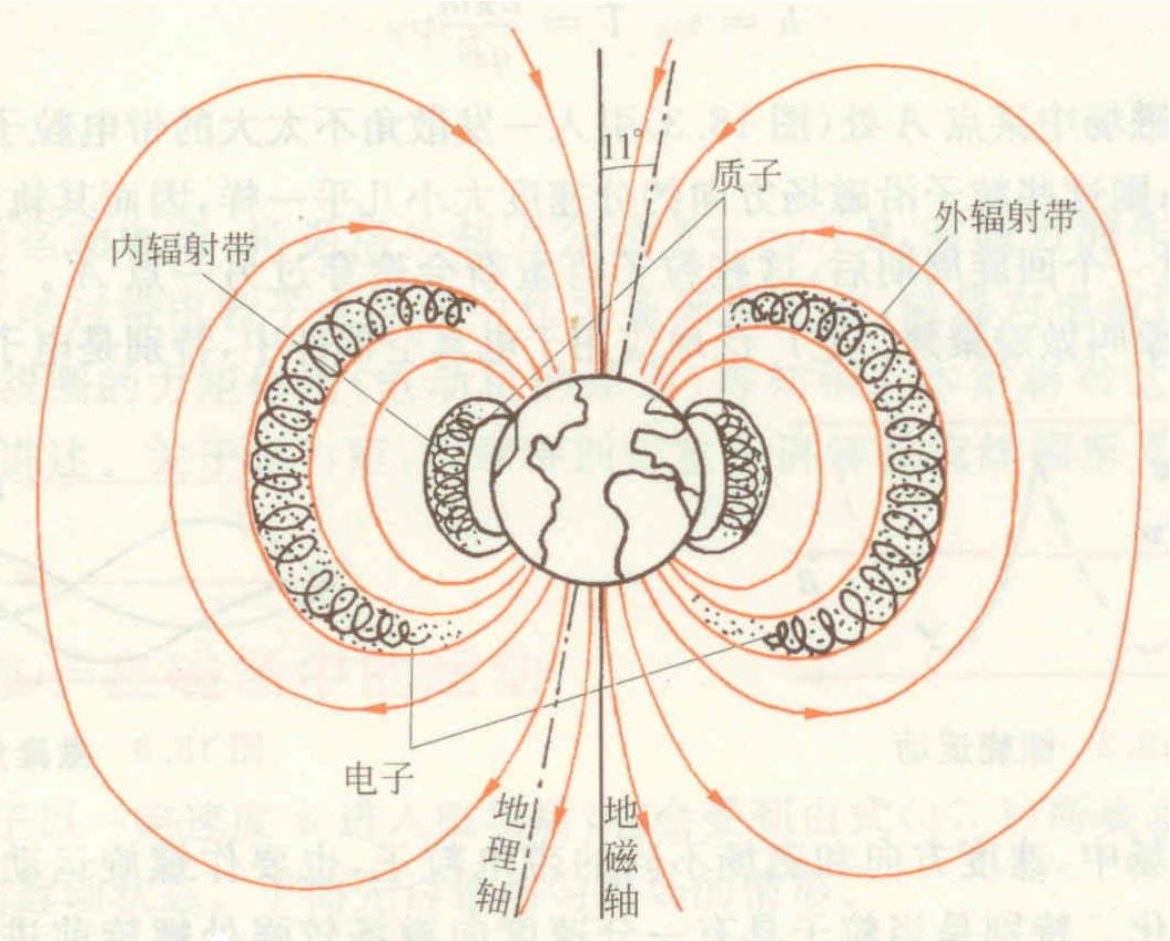
地球的范艾仑辐射带

内带

主要质子

外带

主要电子



在地磁两极附近，由于磁感线与地面垂直，外层空间入射的带电粒子可直接射入高空大气层内，它们和空气分子碰撞产生的辐射，就形成了极光。

18.2 电场和磁场的相对性和统一性

$$\boxed{\vec{f} = q\vec{E} + q\vec{\upsilon} \times \vec{B}}$$ 洛仑兹力公式

某个瞬间参考系速度与电荷运动速度相同

磁力消失? 在不同参考系物理结果应相同

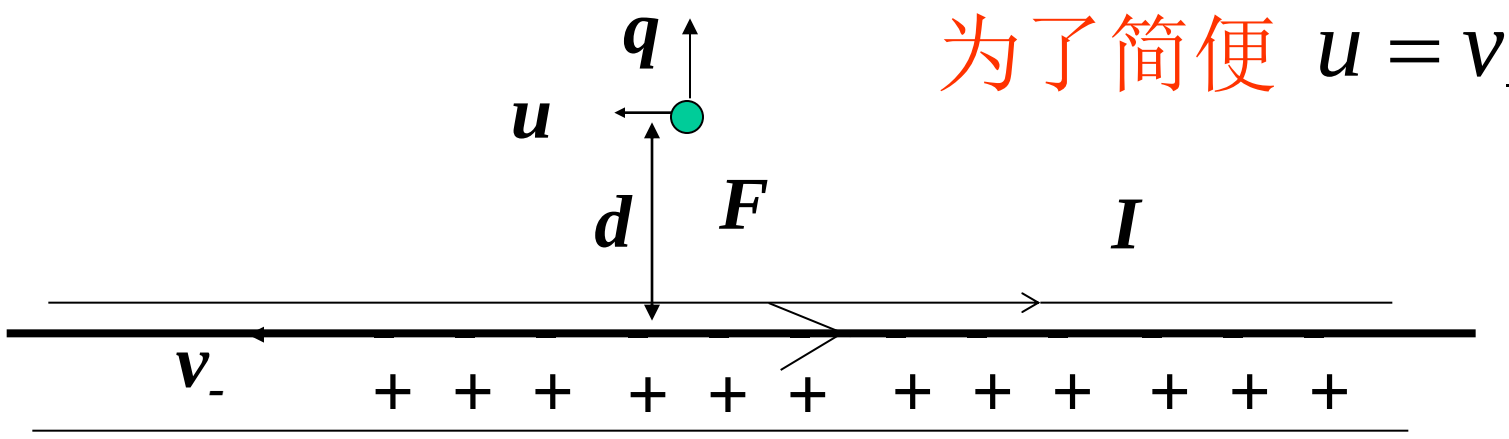
相对论角度

磁场在另一个惯性系变成电场

电场在另一个惯性系变成磁场

无限长载流导线电流 I 与一运动电荷 q 相距 d

为了简便 $u = v_-$



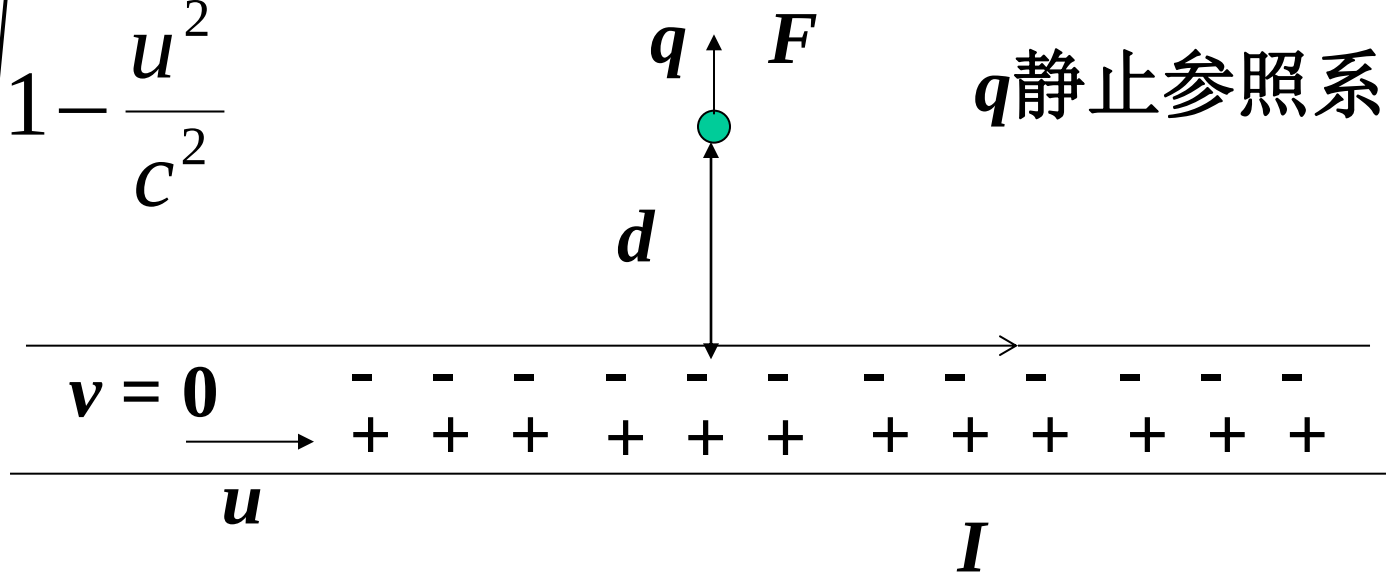
I 在 q 处 B 为

$$B = \frac{\mu_0 I}{2\pi d}$$

根据洛伦兹力，q受力

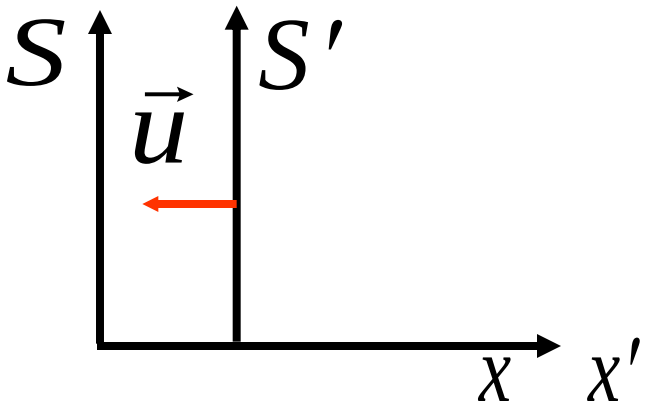
$$F_m = \frac{\mu_0 I q u}{2\pi d}$$

$$\gamma = \frac{1}{\sqrt{1-\frac{u^2}{c^2}}}$$



$$\because l = \frac{l_0}{\gamma} \qquad l_+ = \frac{l_{+0}}{\gamma}$$

$$l_- = \gamma l_{-0}$$



电荷密度反比于间距

$$\lambda_- = \frac{\lambda_0}{\gamma} \qquad \lambda_+ = \lambda_0 \gamma$$

$$F' = q\frac{\lambda_+ - \lambda_-}{2\pi\varepsilon_0 d} = \frac{\lambda_0}{2\pi\varepsilon_0 d}\gamma\frac{u^2}{c^2}q = \frac{\mu_0 I q u}{2\pi d}\gamma$$

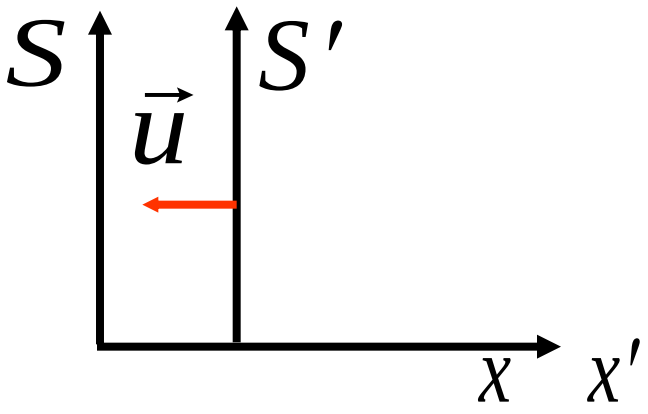
$$\boxed{F' = \gamma F}$$

$$\frac{F_y'}{F_y} = \frac{\Delta p_y'/\Delta t'}{\Delta p_y/\Delta t}$$

$\searrow \gamma$

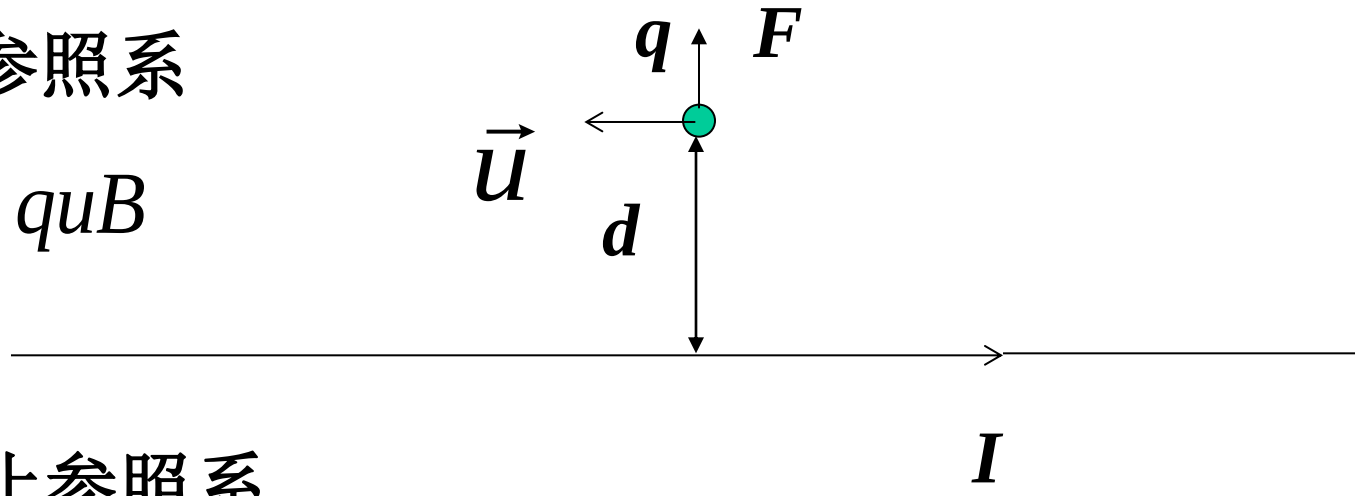
原时最短

$$\boxed{\Delta t = \gamma \Delta t'}$$



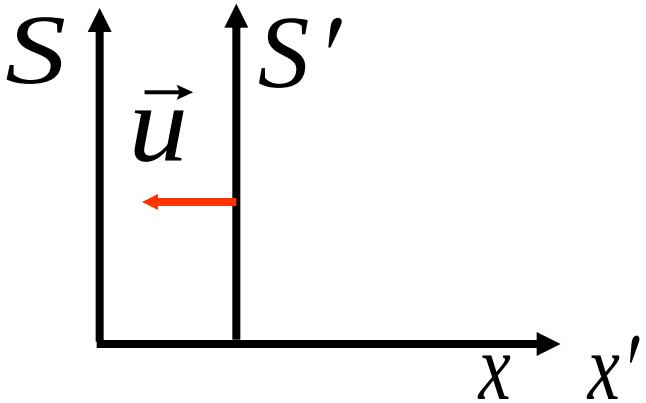
地面参照系

$$F = quB$$



q静止参照系

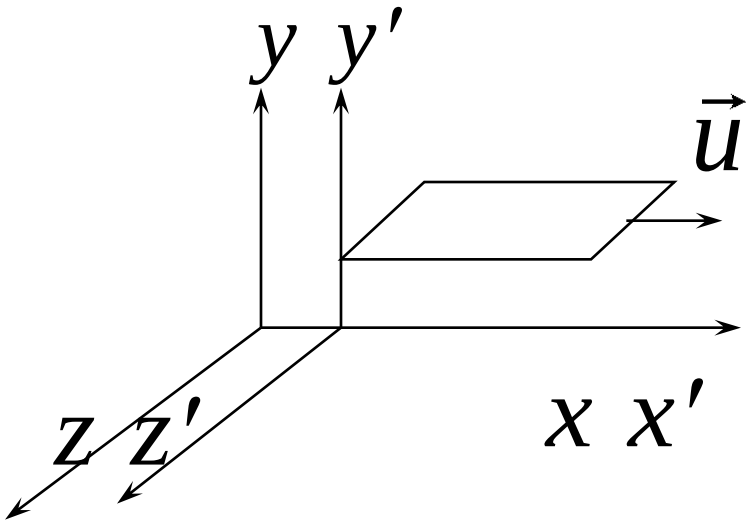
$$\gamma F = \gamma quB \qquad E'_y = \gamma uB = -\gamma u_x B$$



地面参考系只有磁场

运动参考系还有电场

均匀带电的无限大平面电荷面密度 σ



S` 系中只有静电场

$$\vec{E}' = \vec{E}'_y = \pm\frac{\sigma}{2\varepsilon_0}\hat{y}$$

$$S: \qquad \vec{j} = \sigma'\vec{u} = \gamma\sigma\vec{u}$$

$$\vec{B} = \pm\frac{\mu_0}{2}\gamma\sigma u\hat{z}$$

$$E_y = \gamma E'_y$$

$$\boxed{\vec{B} = \frac{\vec{u}\times\vec{E}}{c^2}}$$

$$= \pm\gamma\frac{E'_y u}{c^2}\hat{z}$$

电场—磁场似乎是完全不同

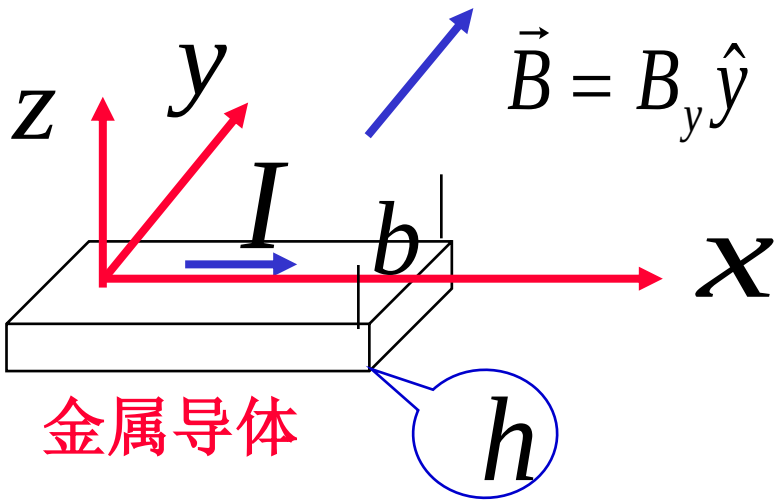
相对论的角度　电场—磁场相互紧密关联

惯性系变换可以使 电场—磁场 互相变换

电磁场方程是在惯性系成立

场线只是形象描述, 不能真把它当“线”
以为可以识别是否相对场运动

18.3　霍耳效应



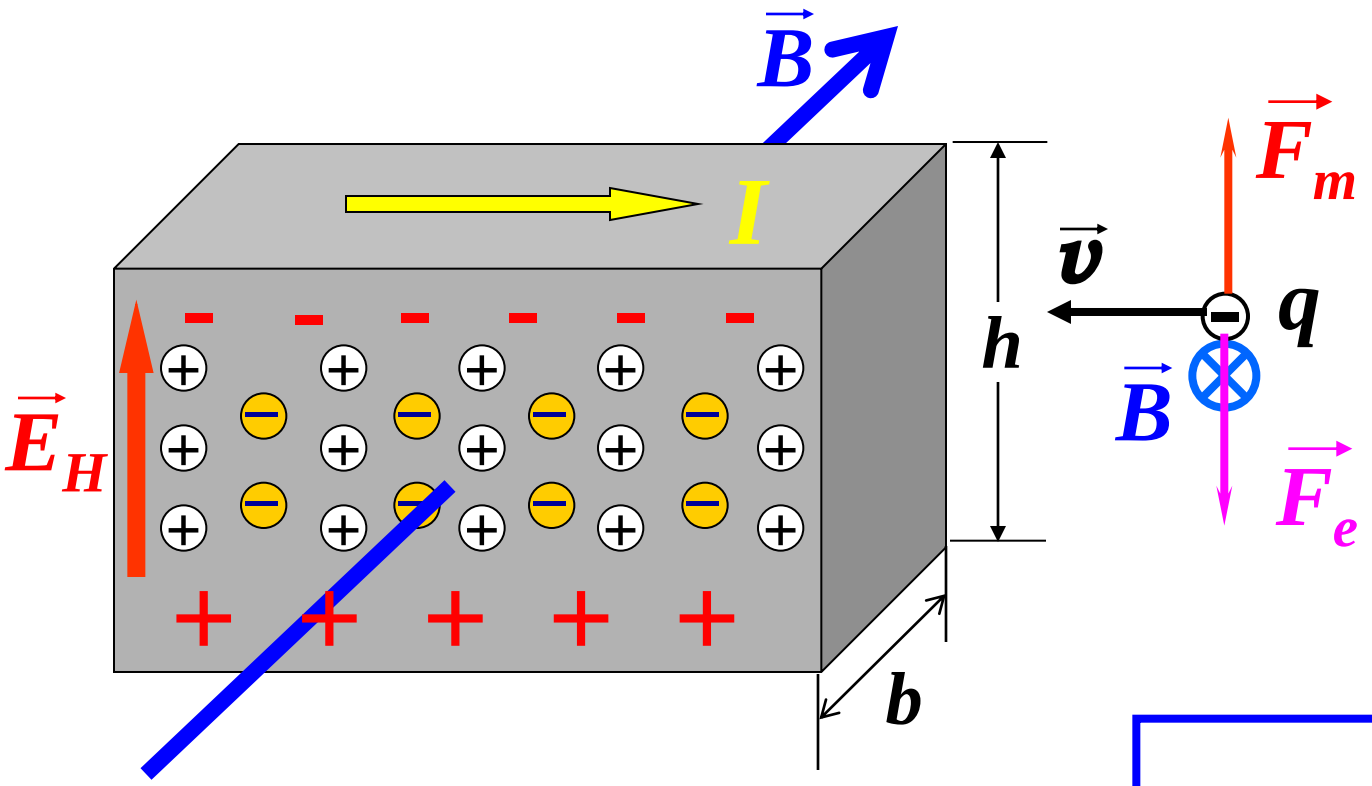
1879年美国物理学家霍耳发现：

对应图中沿 z方向有电势差

霍耳电势差：

$$U_H \propto \frac{IB}{b}$$

霍耳电势差：



$$q\upsilon B = qE_H$$

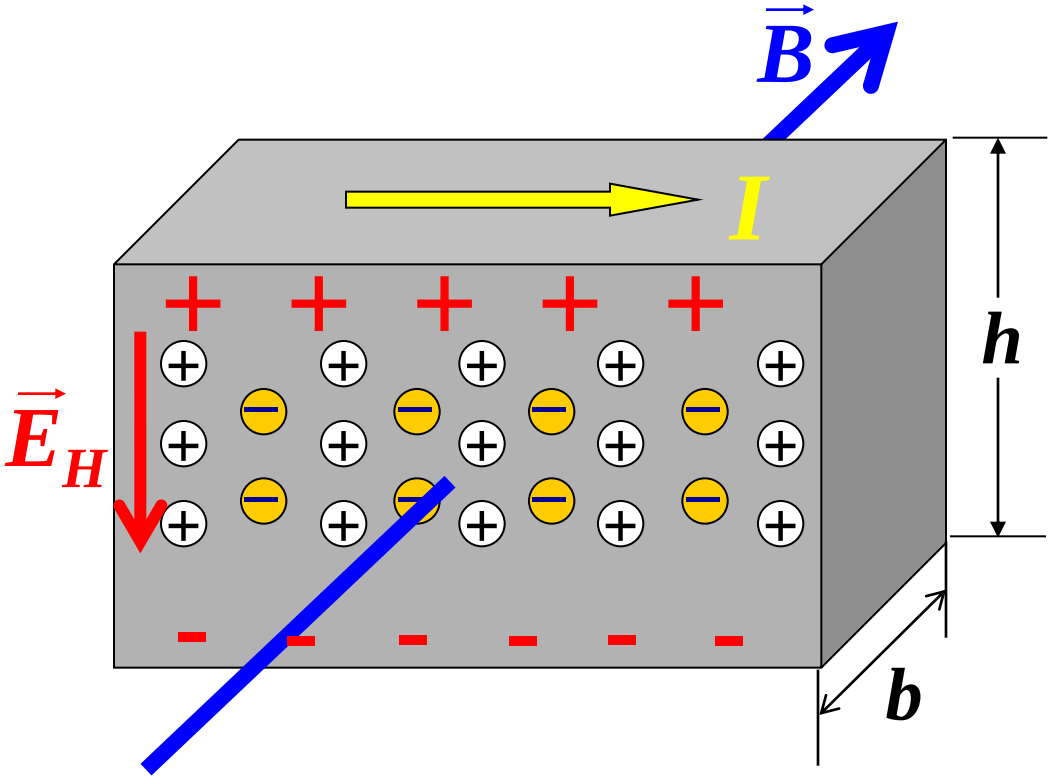
$$nq\upsilon bh = I$$

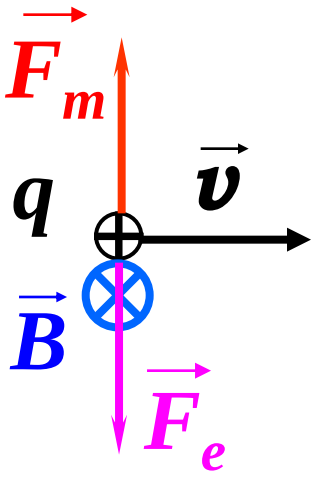
$$E_H h = U_H$$

$$\boxed{K_H = \frac{1}{nq}}$$

$$U_H = \frac{IB}{nbq} = K_H \frac{IB}{b}$$

K_H —— 霍耳系数





$$U_H = \frac{IB}{nbq} = K_H \frac{IB}{b}$$

判断载流子种类

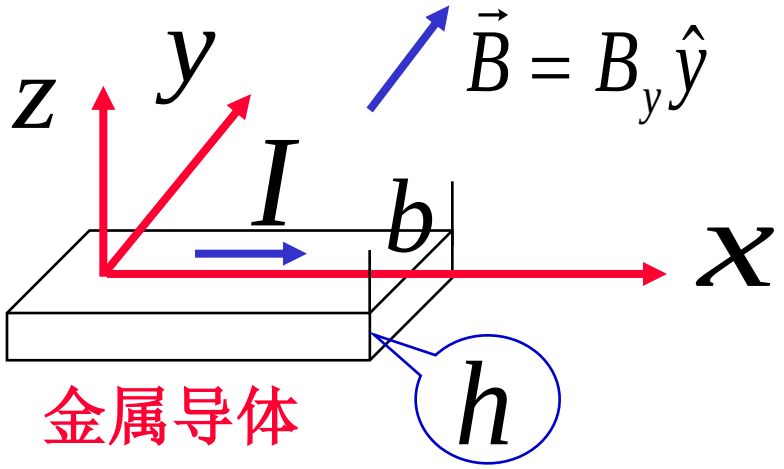
测量载流子浓度　　测磁场等

霍耳电阻 $$R_H = \frac{U_H}{I} = \frac{B}{nqb} \propto B$$

霍耳效应

精确的解释只能用电子的量子理论

（包括反常情形）



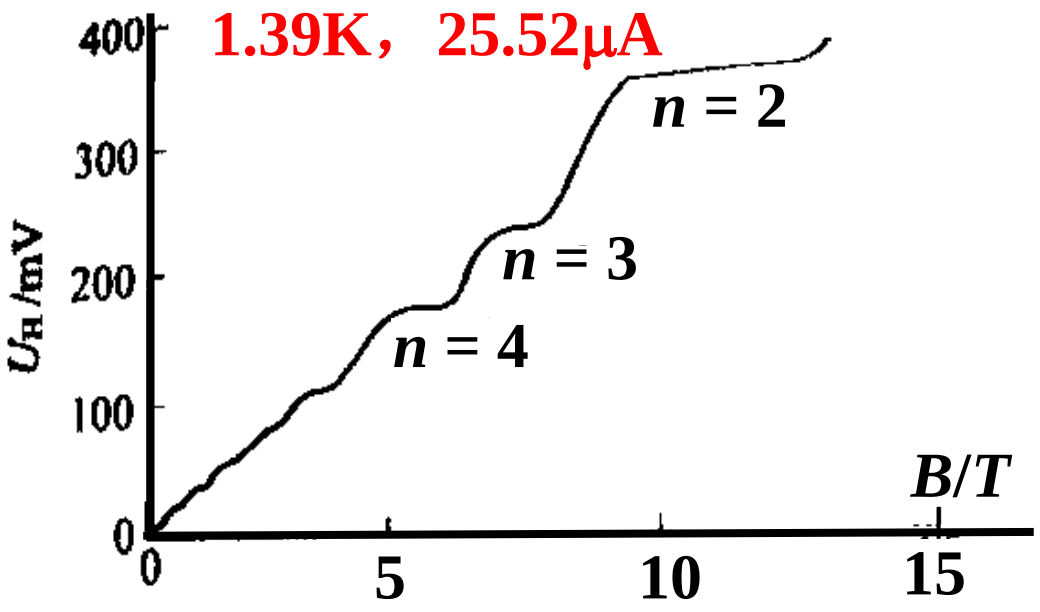
*量子霍耳效应：

1980年克里青发现，

在极低温、强磁场下

$$R_H \not\propto B$$

$$R_H = \frac{R_K}{n}, \quad n = 1,2,3,\cdots$$



克里青（Klitzing）常量 $R_K = \frac{h}{e^2} = 25812.80\Omega$

R_K的测量准确到10^{-10}

1990年定义 $1\Omega = \frac{R_K}{25812.80}$

1985年诺贝尔物理学奖

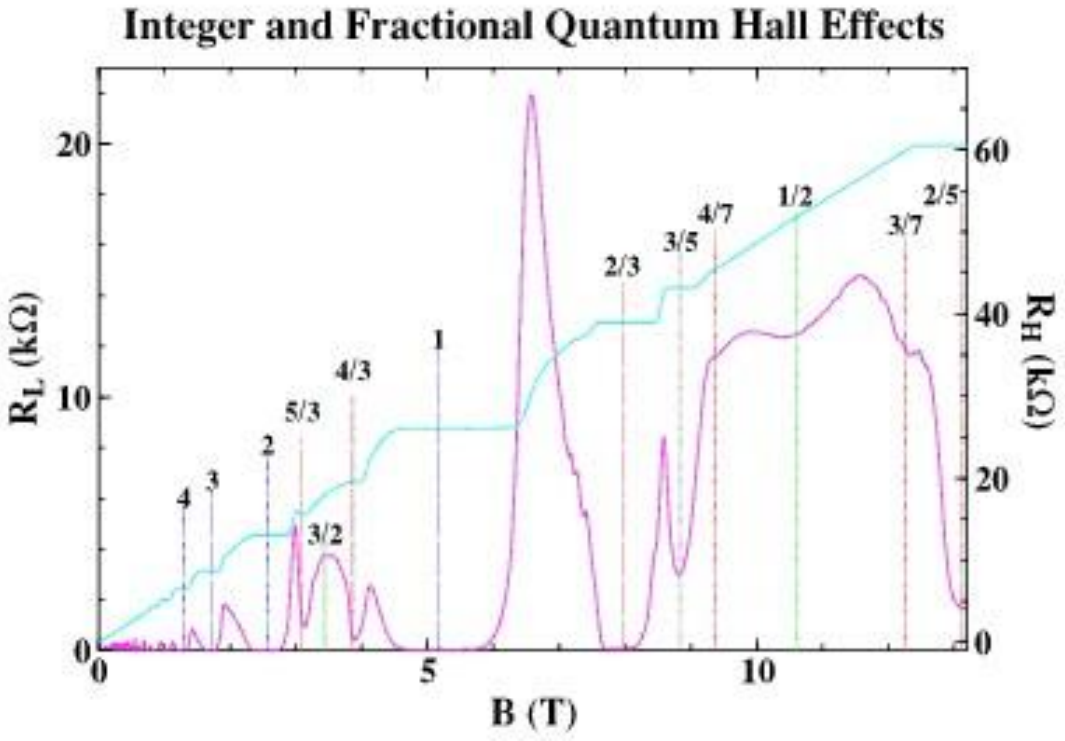
*分数量子霍耳效应：

崔琦和施特默（Störmer）发现在更强的磁场下，*n* 可以是分数，如：1/3、1/5、1/2、1/4等，这称为分数量子霍耳效应。

劳克林（Laughlin）：携带分数电荷的准粒子

劳克林、施特默和崔琦

1998年诺贝尔物理学奖

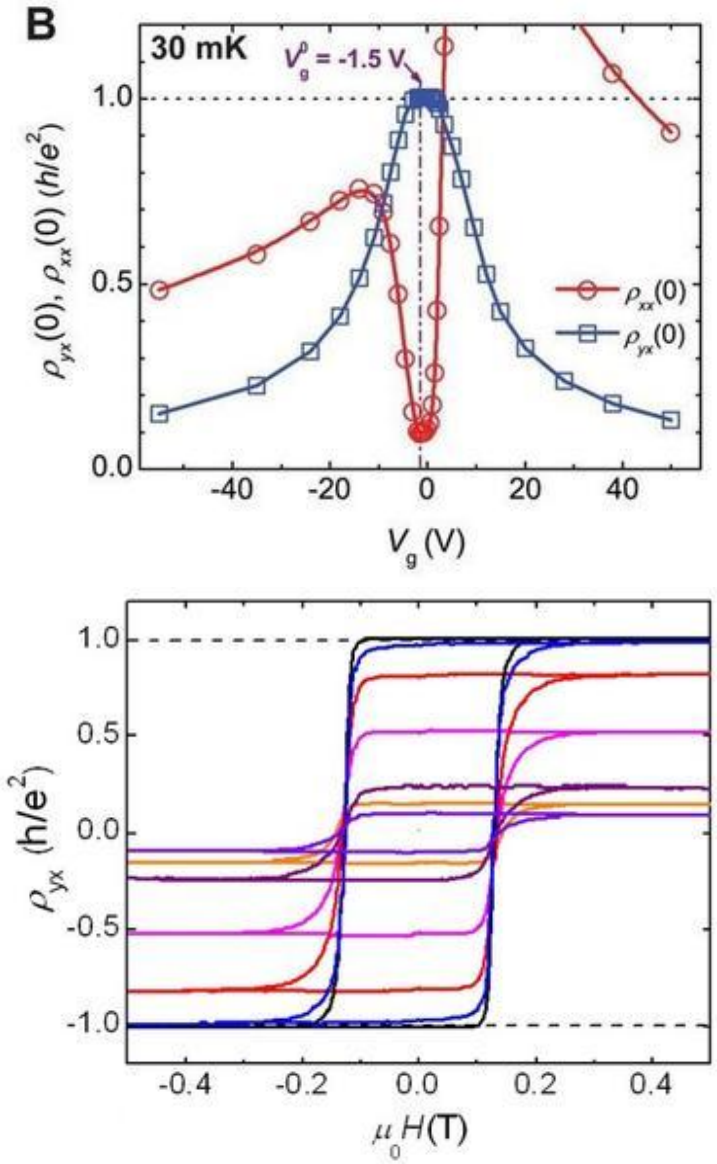


反常霍耳效应：

1880年霍耳发现, 不需要外加磁场, 某些材料靠自发磁化产生霍耳效应.

量子反常霍耳效应：

2013年3月15日，《科学》（Science）杂志在线发文, 清华大学物理系薛其坤等与科学院物理所及斯坦福大学物理学家合作, 观察到量子反常霍耳效应.



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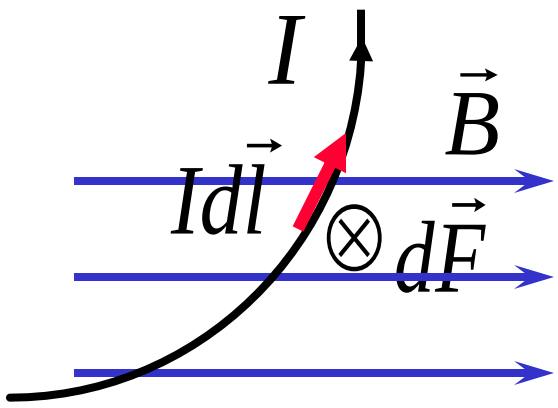
18.4 安培力



怎么计算电流受到的磁场力?

安培指出 任意电流元受力为

安培力公式 $\boxed{d\vec{F} = Id\vec{l} \times \vec{B}}$

整个电流受力 $$\boxed{\vec{F} = \int_{(l)} Id\vec{l} \times \vec{B}}$$

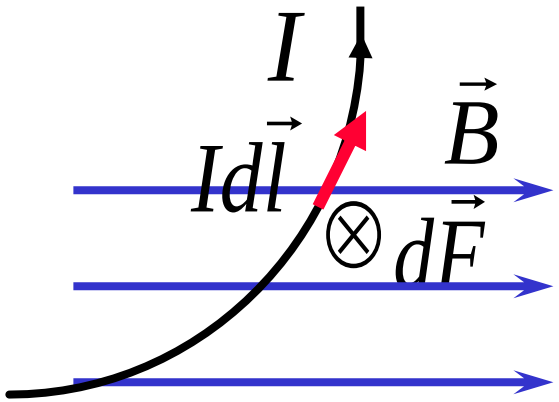
- 安培力的本质是洛仑兹力

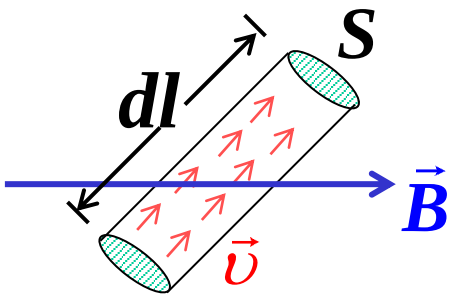
$$d\vec{F} = Id\vec{l} \times \vec{B}$$

$$= nq\vec{\upsilon}\Delta S dl \times \vec{B}$$

$$= \vec{j} \times \vec{B} dV$$

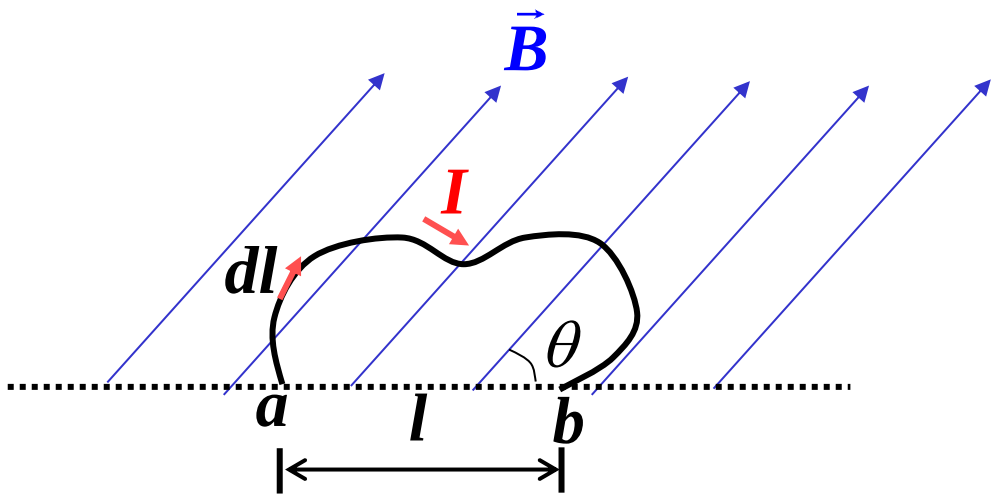
$$= dNq\vec{\upsilon} \times \vec{B}$$





载流子受到的磁力传递给导线本体

例1 在均匀磁场 $\vec{B}$ 中有一段弯曲导线 ab 通有电流 I 求此段导线受到的磁力



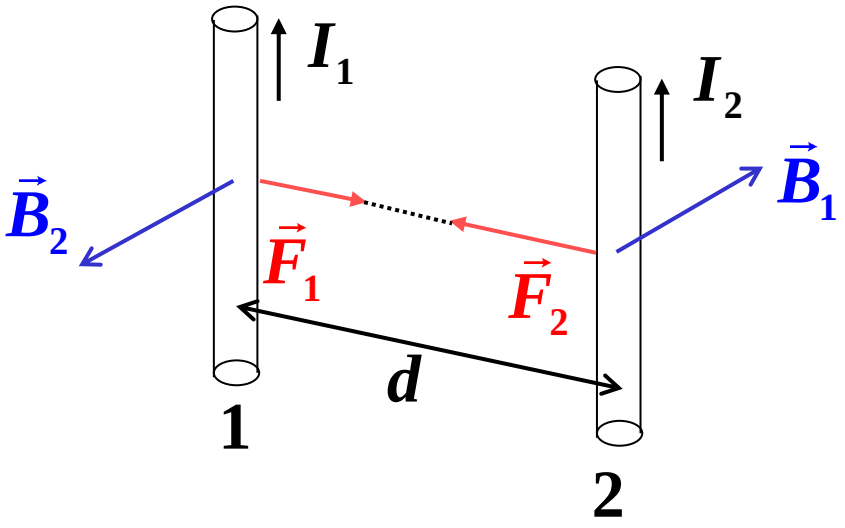
解 $\vec{F}=\int_a^b Id\vec{l}\times\vec{B}=I(\int_a^b d\vec{l})\times\vec{B}=I\vec{l}\times\vec{B}$

$F=IlB\sin\theta$ 方向垂直纸面向外

推论

在均匀磁场中的闭合载流回路整体上不受安培力

平行导线间的相互作用力



设导线直径远小于导线间距离？

导线1在导线2处产生的磁场 $B_1 = \dfrac{\mu_0 I_1}{2\pi d}$

导线2单位长度上受到的力 $F_2 = I_2 B_1 = \dfrac{\mu_0 I_1 I_2}{2\pi d}$

同理，导线1单位长度上受到的力 $F_1 = \dfrac{\mu_0 I_1 I_2}{2\pi d}$

电流方向相同，相吸，电流方向相反，相斥 （演示实验）

电流强度的单位　安培 A

设在真空中两根无限长的平行直导线相距$1m$，通以大小相同的恒定电流，如果导线每米长度受到的作用力为 $2\times10^{-7}N$，则每根导线中的电流强度就规定为 $1A$

$$F=\frac{\mu_0 I_1 I_2}{2\pi d}\Rightarrow \mu_0=\frac{2\pi dF}{I_1 I_2}=\frac{2\pi\times1\times2\times10^{-7}}{1\times1}$$

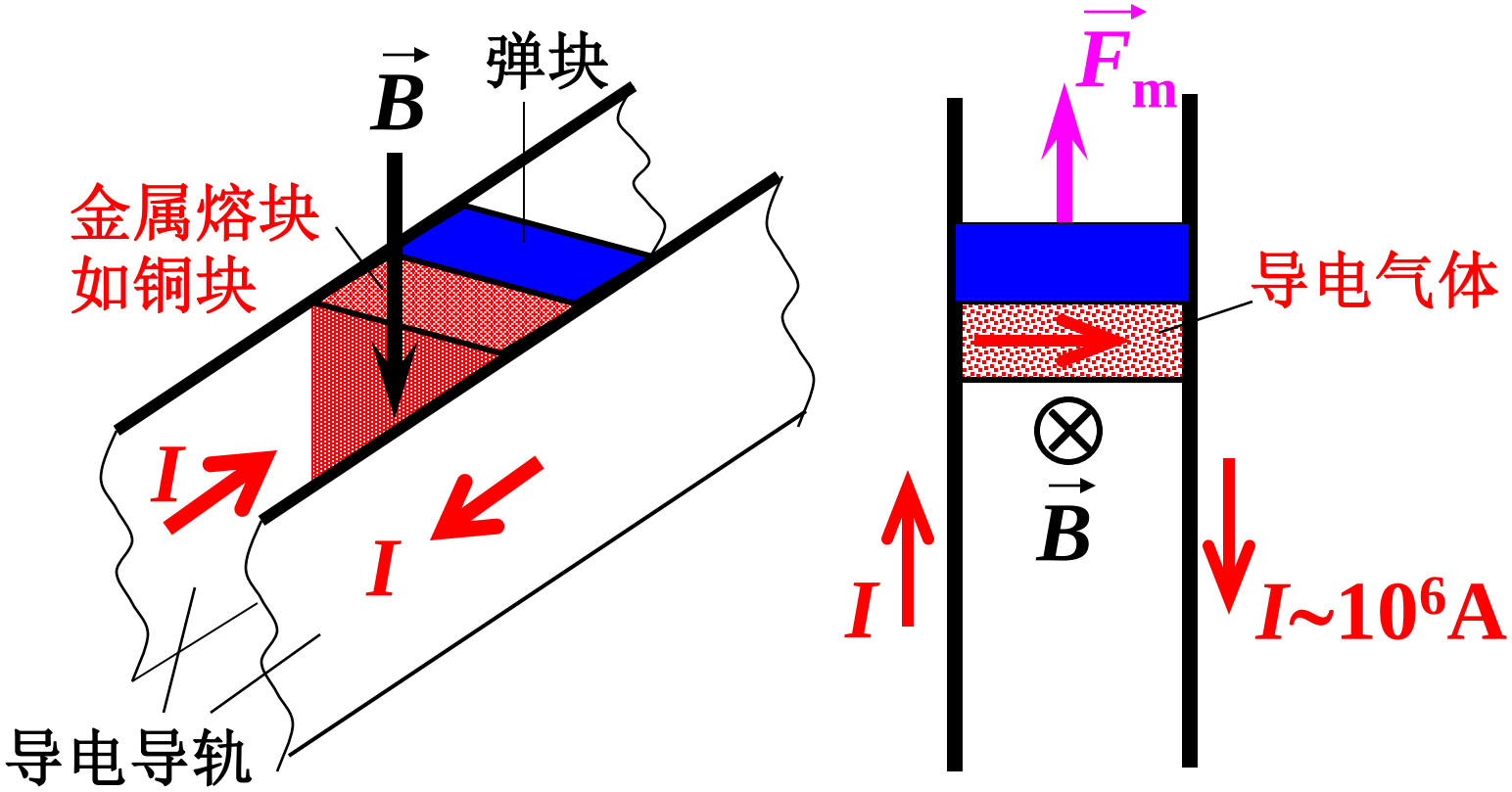
$$=4\pi\times10^{-7}(N/A^2)$$

电量的单位 库仑 C

在通有1A电流的导线中，每秒钟流过导线任一横截面上的电量

磁流体船

电磁轨道炮



弹块 $\bar{a} \sim 10^6 g$，在1ms内，速度可达10km/s

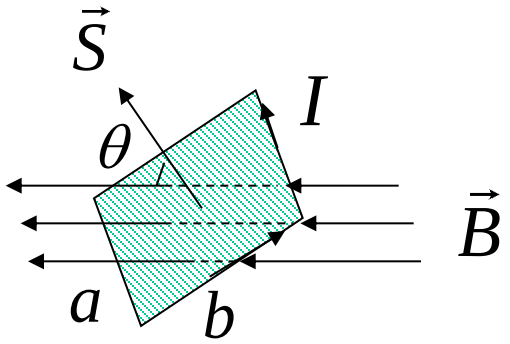
18.5 载流线圈在磁场

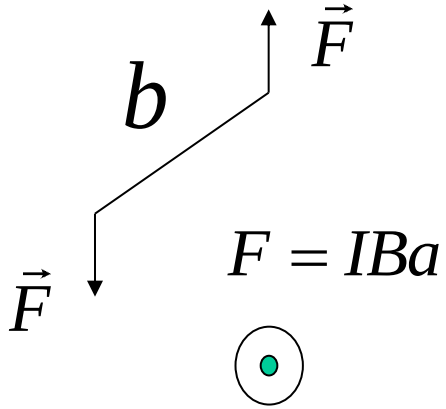
载流线圈在均匀磁场中

- 合力 $\boxed{\vec{F}_{合} = 0}$

$$\vec{F} = \int_{(l)} I d\vec{l} \times \vec{B}$$

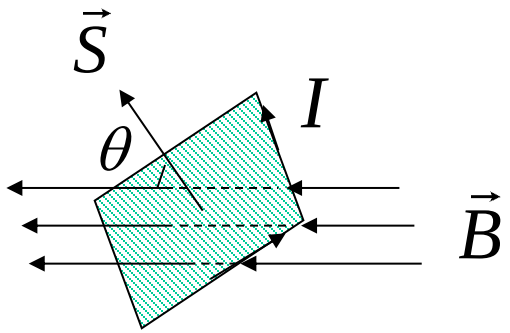
- 力矩 $\boxed{\vec{M} = \vec{p}_m \times \vec{B}}$





$$M = Fb\sin\theta = IBab\sin\theta = p_m B\sin\theta = \vec{p}_m \times \vec{B}$$

载流线圈在均匀磁场中的能量



I 不变

$$W_m(\theta)-W_m(0)=\int_0^{\theta} M d\theta$$

$$=\int_0^{\theta} ISB\sin\theta d\theta$$

$$=-ISB(\cos\theta-\cos 0)$$

选择θ=π/2时为势能零点：$W_m(\theta)=-ISB\cos\theta$

$$\boxed{W_m=-\vec{p}_m\cdot\vec{B}}$$

	力矩	势能
电偶极矩	$\boxed{\vec{M} = \vec{p}_e \times \vec{E}}$	$\boxed{W_e = -\vec{p}_e \cdot \vec{E}}$
磁矩	$\boxed{\vec{M} = \vec{p}_m \times \vec{B}}$	$\boxed{W_m = -\vec{p}_m \cdot \vec{B}}$

磁矩在磁场中的受力

磁矩在磁场中的势能

$$W = -\vec{p}_m \cdot \vec{B}$$

磁矩在磁场中的受力

$$\vec{F} = -\nabla W = \nabla(\vec{p}_m \cdot \vec{B})$$

$$= \nabla(p_{mx}B_x + p_{my}B_y + p_{mz}B_z)$$

$$= p_{mx}\nabla B_x + p_{my}\nabla B_y + p_{mz}\nabla B_z$$

如果磁场只有 x 方向分量 B_x

$$\vec{F} = p_{mx}\nabla B_x = p_{mx}\left(\frac{\partial B_x}{\partial x}, \frac{\partial B_x}{\partial y}, \frac{\partial B_x}{\partial z}\right)$$

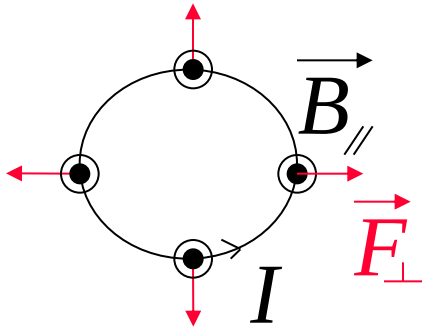
*载流线圈在非均匀磁场中受力

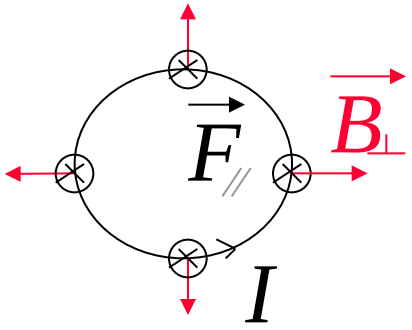
演示实验

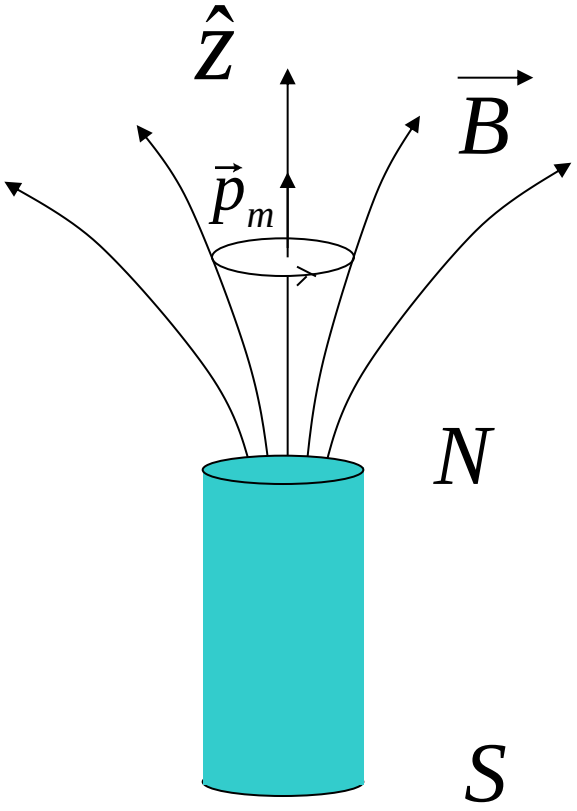
I 不变

$$\vec{B} = \vec{B}_{//} + \vec{B}_{\perp}$$

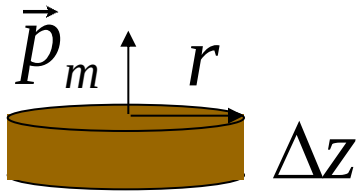
$$\vec{p}_m = I\Delta S\ \hat{z}$$







$$\vec{F}_{\perp} = 0 \qquad \vec{F}_{//} = -2\pi r I B_{\perp}\hat{z}$$



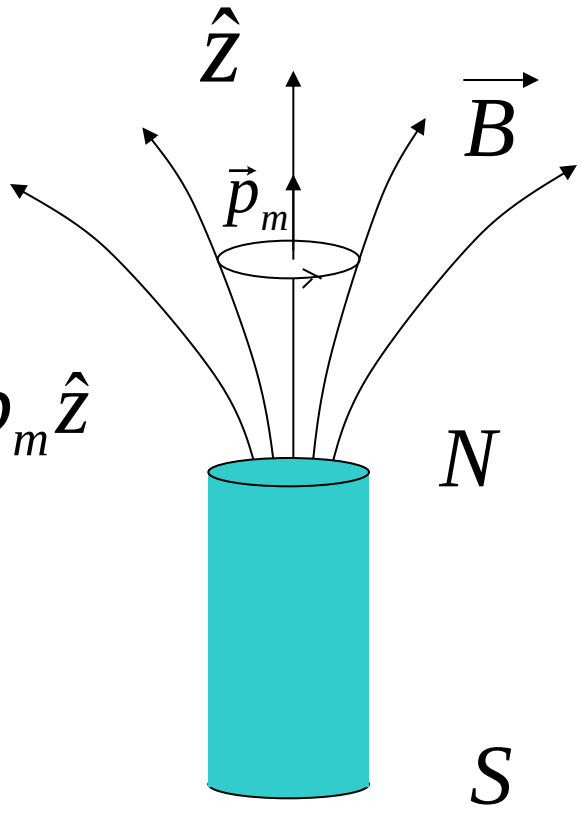
$$\oiint_S \vec{B}\cdot d\vec{S} = 0 \quad \Rightarrow \quad [B_{//}(z+\Delta z) - B_{//}(z)]\Delta S + B_{\perp}2\pi r\Delta z = 0$$

$$[B_{//}(z+\Delta z)-B_{//}(z)]\Delta S + B_{\perp}2\pi r\Delta z = 0$$

$$2\pi r B_{\perp} = -\frac{\partial B_{//}}{\partial z}\Delta S$$

$$\vec{F}_{//} = -2\pi r I B_{\perp}\hat{z} = \frac{\partial B_{//}}{\partial z} I\Delta S\hat{z} = \frac{\partial B_{//}}{\partial z} p_m \hat{z}$$

$$F_z\hat{z} = p_m \frac{\partial B_{//}}{\partial z}\hat{z} = \vec{p}_m \cdot \vec{\nabla} B_z \hat{z}$$



$$\boxed{\vec{F} = \vec{p}_m \cdot \vec{\nabla}\vec{B}}$$